GCNGrasp-VP: Affordance-Guided View Planning for Efficient Task-Oriented Grasping

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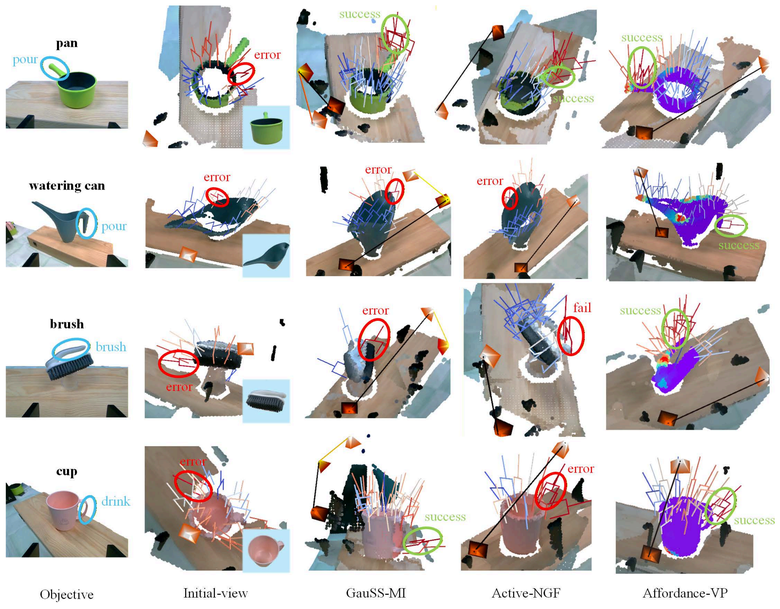
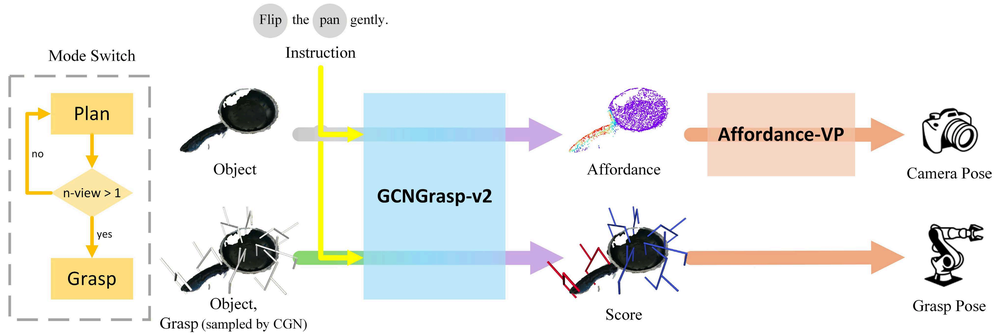
01 Why move the camera?

Occlusion hides regions needed by the instruction. A useful new view should reveal **task-relevant grasp evidence**, not just uncertain geometry.



02A Use task affordance to choose the next view

HOW · PLANNER



GCNGrasp-VP = GCNGrasp-v2 model + Affordance-VP planner. Predict task affordance, move once, then re-score grasps.

Model: GCNGrasp-v2

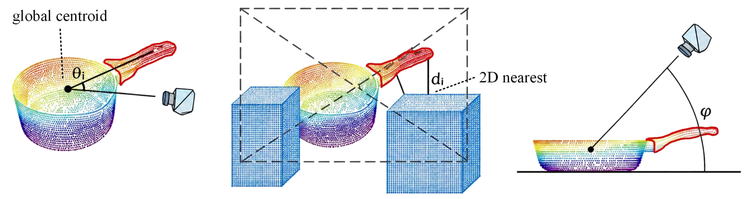
One object–task representation supports both candidate-grasp scoring and per-point affordance prediction.

Core insight

Affordance can guide where to look next—not only which visible grasp to choose.

Planner: Affordance-VP

Cluster the strongest task evidence, then select a feasible view that aligns with the target and increases clearance from occluders—without reconstructing the full scene.



VIEW OBJECTIVE. Favor direct target alignment (small θ_i), large 2-D clearance from the nearest occluder (d_i), and a non-extreme elevation (φ).

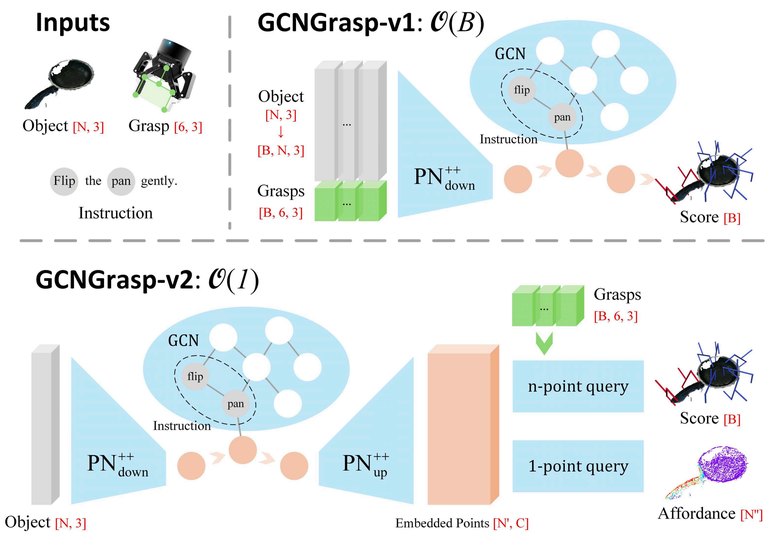
How the planner ranks views

> **Target.** The largest cluster among the top 10% affordance points identifies the task-relevant region to reveal.

1. **Align.** Look directly toward the target region.
2. **Clear.** Prefer larger projected distance from nearby occluders.
3. **Elevate.** Penalize extreme top-down camera poses.
4. **Select.** Choose the feasible sampled pose with the lowest weighted loss.

02B Share task features across grasps

HOW · MODEL

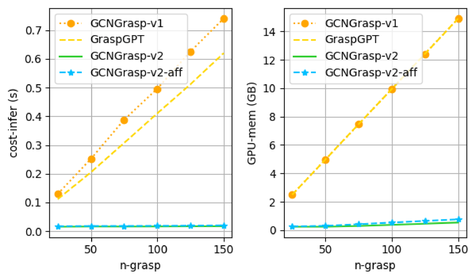


ARCHITECTURE DIFFERENCE. GCNGrasp-v1 repeats object–task encoding for each of *B* candidate grasps, so evaluation grows with *B*. GCNGrasp-v2 encodes shared object–task features once, then reuses them to query all grasps and predict affordance, making candidate-grasp evaluation constant with respect to *B*. *B* is the number of candidate grasps.

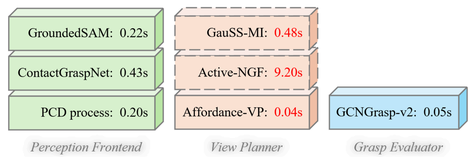
03 Does one task-aware view help?

EVIDENCE

One planned view update is enough for Affordance-VP to achieve the **highest success rate on all four real-world tasks.**



EFFICIENT AT SCALE. GCNGrasp-v2 avoids linear time and memory growth as *B* increases.



PRACTICAL ACTIVE PERCEPTION. Task-aware view selection adds little pipeline overhead.

Task-oriented grasp prediction

mAP (%)

n	Method	Pan pour	Watering can dispense	Brush brush	Cup drink
1	initial-view	39.27	46.70	19.56	14.67
2	GauSS-MI	76.59	65.29	50.49	33.35
	Active-NGF	82.95	67.89	50.83	22.21
	Affordance-VP	**98.42**	**70.28**	**76.10**	50.16
3	GauSS-MI	**89.92**	62.91	34.79	**51.36**
	Active-NGF	84.62	66.03	62.45	38.62
	Affordance-VP	78.33	**76.52**	**86.54**	**57.04**

Real-world grasp execution

successes / trials

n	Method	Pan pour	Watering can dispense	Brush brush	Cup drink
1	initial-view	2/24	7/20	4/28	2/24
2	GauSS-MI	17/24	10/20	6/28	6/24
	Active-NGF	19/24	11/20	11/28	2/24
	Affordance-VP	**24/24**	**14/20**	**20/28**	**10/24**

n = total observed views; *n* = 2 means one additional view. **Best** / **second-best** per task among compared rows.

Why task-aware views help

Baselines prioritize scene uncertainty; Affordance-VP observes regions that support the instruction.

The first update delivers the main gain

At *n* = 2, unweighted mean mAP rises **30.05 → 73.74** (+43.69 pp); pooled successes rise **15/96 → 68/96**.

More views do not help uniformly

At *n* = 3, pan-pour falls 98.42 → 78.33. The paper points to fusion noise; severe occlusion can also misplace affordance peaks.

TAKEAWAY

Use task affordance to decide where to look next.

GCNGrasp-VP couples efficient grasp-and-affordance prediction with task-aware view selection in one active loop.

ACKNOWLEDGEMENT

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Code & models

github.com/Instinct323/GCNGrasp-VP